Differential Geometry: midsem exam

Aditya Padiyar

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Definition 1. *A submanifold S is called totally geodesic if for every $p \in S$ and $v \in T_pS$ we have range $\gamma_v \subseteq S$*

Definition 2. *We say that $X \in \mathcal{X}(M)$ is compatible with S if $X_{|S} \in \mathcal{X}(S)$. This space is $\mathcal{X}_S(M)$*

Claim 1. *If $X, Y \in \mathcal{X}_S(M)$ then $[X,Y]_{|S} = [X_{|S}, Y_{|S}]_S$*

1 Second fundamental form

Fix a submanifold S. Let ∇ and $\tilde{\nabla}$ denote the Levi-Civita connections on M and S respectively.

Definition 3. *Define $\mathcal{F} \in B(\mathcal{X}_S(M), C^\infty(S, \mathbb{R}^k))$ by*

$$X, Y \to (\nabla_X Y) - \tilde{\nabla}_X Y$$

Claim 2. *range $\mathcal{F} \subseteq \mathcal{Y}(S)$*

Proof. Observe that $[\tilde{\nabla}_X Y](q) = \tilde{\pi}([\nabla_X Y](q))$ □

Claim 3. *$\mathcal{F}$ is symmetric*

Claim 4. *$\mathcal{F}(fX, Y) = f\mathcal{F}(X,Y)$ for any $f \in C^\infty(M)$*

Claim 5. *If $X(p) = 0$ then $\mathcal{F}(X,Y)(p) = 0$*

Proof. Choose a tangential frame $X_i \in \mathcal{X}(U)$ on some neighborhood U. We have $X = \sum f_i X_i$ with $f_i(p) = 0$. Now use the bilinearity with respect to smooth functions. □

Claim 6. *Suppose $v \in T_pS$. Then there is a neighborhood U of p and $X \in \mathcal{X}_{S\cap U}(U)$ such that $X(p) = v$*

Proof. Let $\phi : \Omega \to U$ be a local parametrization compatible with S, with inverse ψ. Let $\varphi : W \to S \cap U$ is the associated local parametrization of S. Now $v = (D\varphi)_0 y$. Define $X : U \to \mathbb{R}^k$ by $X(q) = (D\psi)_q^{-1}(y, 0)$ □

Claim 7. *Suppose $v_1, v_2 \in T_pS$ and $X_i, Y_i \in \mathcal{X}_S(M)$ are such that*

$$X_i(p) = Y_i(p) = v_i$$

Then $\mathcal{F}(X_1, X_2)(p) = \mathcal{F}(Y_1, Y_2)(p)$

Proof. We have $\mathcal{F}(X_1, X_2) - \mathcal{F}(Y_1, Y_2) = \mathcal{F}(X_1 - Y_1, X_2) - \mathcal{F}(Y_1, X_2 - Y_2)$. But $X_1 - Y_1$ and $X_2 - Y_2$ vanish at p □

Definition 4. *We have $Q_p \in B(T_pS, N_pS)$ defined by $(v, w) \to \mathcal{F}(X, Y)(p)$*

Claim 8. *Q_p is symmetric*

Claim 9. *Suppose γ is a smooth curve in S defined on an interval I and V is a smooth vector field along γ satisfying $V(t) \in T_{\gamma(t)}S$. Then we have*

$$(cov\ V)(t) = (\widetilde{cov}\ V)(t) + Q_{\gamma(t)}(\gamma'(t), V(t))$$

where cov and $\widetilde{cov}$ denote the covariant derivatives along γ in M and S respectively.

Proof. Consider the following

- Choose t_0 and a compatible local parametrization $\phi : \Omega \to U$
- The associated local parametrization of S is $\varphi : W \to S \cap U$
- Let $\gamma(t_0) = p$ and $T : \mathbb{R}^k \to N_pS$ be the orthogonal projection
- $V(t) = V_i(t)\varphi_i(\gamma(t))$ and $\gamma(t) = \varphi(x(t))$
- $Q_p(\gamma'(t_0), V(t_0)) = \dot{x_i}(t_0)V_j(t_0)Q_p(\varphi_i(p), \varphi_j(p))$
- $\dot{V}(t_0) = \dot{x_i}(t_0)V_j(t_0)(D_{\varphi_i}\varphi_j)(p) + \dot{V_j}(t_0)\varphi_j(p)$
- $(\text{cov} - \widetilde{\text{cov}})V(t_0) = T(\dot{V}(t_0)) = \dot{x_i}(t_0)V_j(t_0)T(D_{\varphi_i}\varphi_j)(p)$
- Now use the definition of Q_p. The required vector field extensions are ϕ_i and ϕ_j respectively

□

Claim 10. *If S is totally geodesic then $Q_p = 0$ for all $p \in S$*

Proof. Let $v \in T_pS$. Applying the previous claim to γ_v we get

$$\widetilde{\text{cov}}\gamma'(0) + Q_p(v, v) = 0$$

Since the first term is in T_pS and the second term is in N_pS we get that $Q_p(v, v) = 0$. Since Q_p is symmetric it must be zero. □

2 Fixed points

Suppose $\mathcal{A} \subseteq \mathrm{Iso}(M)$, where $\mathrm{Iso}(M)$ denotes the group of Riemannian isometries on M. Define

$$C = \{q \in M : q \text{ is a fixed point of every } A \in \mathcal{A}\}$$

For $q \in C$ define the subspace W_q of T_qM to be the points which are fixed by each DA_q, for $A \in \mathcal{A}$

Claim 11. *C is closed in M*

Claim 12. *Suppose $q \in C$ and $v \in \mathcal{E}_q \cap W_q$. Then $\exp_q(v) \in C$*

Proof. Let $A \in \mathcal{A}$. We have $A \exp_q(v) = \exp_q(DA_q v) = \exp_q(v)$, where the last equality holds because $v \in W_q$. □

Claim 13. *Let $q \in C$ and $U = \exp_q(B_\rho)$, where ρ is the injectivity radius. Then $C \cap U = \exp_q(W_q \cap B_\rho)$*

Proof. Consider the following:

- RHS is contained in LHS by 12
- If $\exp_q(v) \in C$ with $||v|| < \rho$, then $\exp_q(DA_q v) = A \exp_q(v) = \exp_q(v)$. Since DA_q is an isometry, $v \in W_q$

□

Claim 14. *C is a submanifold of M*

Proof. We have that $W_q \cap B_\rho$ is a submanifold of B_ρ. Since $\exp_q : B_\rho \to U$ is a diffeomorphism, by 13 $C \cap U$ is a submanifold. Since this holds for all $q \in C$ and being a manifold is a local property, C is a manifold. □

Claim 15. *Every connected component of a manifold is a submanifold*

Proof. Every manifold is locally path connected. Now use Theorem 25.4 of Munkres. □

Claim 16. *We have $T_qC = W_q$*

Proof. This follows directly from 13 □

Claim 17. *C is totally geodesic*

Proof. Suppose γ_v is the maximal geodesic at $q \in C$ with $v \in T_qC$. Now $\gamma_v(t) = \exp_q(tv) \in C$, by using 16 and 12 □

Suppose now that C is a connected one dimensional submanifold of M and $\sigma : \mathbb{R} \to C$ is a diffeomorphism

Claim 18. *We have* $(cov\ \sigma')(t) \in W_{\sigma(t)}$ *where the covariant derivative is in* M

Proof. By 9, as well as 10 and 17, the covariant derivative is in C. Now use 16. □

Claim 19. *If* $|\sigma'(t)| = 1$ *then* σ *is a geodesic in* M

Proof. Consider the following:

- Differentiating, we get $\langle \sigma''(t), \sigma'(t) \rangle = 0$
- Hence $\langle (\text{cov}\sigma')(t), \sigma'(t) \rangle = 0$
- But $(\text{cov}\sigma')(t) = f(t)\sigma'(t)$
- Hence $f = 0$

□

3 Uniformly normal neighborhood

Suppose we have that

- W is a uniformly δ-normal neighborhood of p
- $q_0 = \exp_p(v_0) \in W$
- ρ is the injectivity radius at p
- $0 < ||v_0|| < \rho$
- Define $f : W \to [0, \infty)$ by $f(q) = [d_M(q_0, q)]^2$
- Define Ψ to be the inverse of the diffeomorphism

$$\exp_{q_0} : B_\delta \to \exp_{q_0}(B_\delta)$$

Claim 20. f *is smooth*

Proof. We have $f(q) = ||\Psi(q)||^2$ □

Claim 21. $grad_f(p) = -2v_0$

Proof. Consider the following:

- Let $\gamma : [0, 1] \to M$ be the minimizing geodesic from p to q_0
- We have $\gamma(t) = \exp_p(tv_0) = \exp_{q_0}((1-t)w_0)$ where $\Psi(p) = w_0$
- By Gauss lemma, $\text{grad}_f(p) = 2(d\exp_{q_0})_{w_0}(w_0) = -2\gamma'(0) = -2v_0$

□

4 Removal of a point

Suppose M is connected and $p \in M$. Define $N = M \setminus \{p\}$

Claim 22. *N is a connected submanifold*

Proof. Consider the following:

- Since N is open in M it is a submanifold.
- Suppose $N = U_1 \sqcup U_2$ is a separation. Taking closures we get $M = \overline{U_1} \cup \overline{U_2}$. Since M is connected these sets cannot be disjoint. Hence $p \in \overline{U_1} \cap \overline{U_2}$. There is a connected neighborhood V of p such that $V \cap N$ is also connected. Without loss of generality we can assume V is disjoint from U_1, which is a contradiction.

□

Claim 23. *For all $q_1, q_2 \in N$ we have $d_N(q_1, q_2) = d_M(q_1, q_2)$*

Proof. Consider the following:

- Suppose $\gamma : [0, 1] \to M$ is a curve from q_1 to q_2. Let S_r be the geodesic sphere of radius r. Let $t_0(r)$ and $t_1(r)$ be the first and last times at which γ is in S_r.
- Then

  $$d_N(q_1, q_2) \leq d_N(q_1, \gamma(t_0(r))) + d_N(\gamma(t_0(r)), \gamma(t_1(r))) + d_N(\gamma(t_1(r)), q_2)$$

  using the triangle inquality
- We can bound this by

  $$\text{length}(\gamma_{|[0,t_0(r)]}) + \text{diam}(S_r) + \text{length}(\gamma_{|[t_1(r),1]}) \leq \text{length}(\gamma) + \text{diam}(S_r)$$

- Now take the limit as $r \to 0$

□